



AJKD Atlas of Renal Pathology: Sarcoidosis

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Clinical and Pathologic Features

Sarcoidosis is a systemic disease of middle-aged patients, with an incidence of 15-20 per 100,000. African Americans are affected much more commonly than those of European ancestry, with the exception of increased incidence in patients of Scandinavian ancestry; there is a slight female preponderance. Organs most commonly involved include lung, lymph nodes (particularly hilar), eyes, and skin. Kidney disease develops in up to a third of patients, and may present as acute kidney injury related to hypercalcemia, or GFR decline and low-level proteinuria due to granulomatous parenchymal injury. Rarely, patients have associated membranous nephropathy, and then have marked proteinuria.

Light microscopy: Numerous interstitial noncaseating well-formed granulomas with multinucleated giant cells are present, and variable interstitial fibrosis and tubular atrophy. Calcifications may be present.

Immunofluorescence microscopy: Noncontributory (unless associated with membranous nephropathy, in which case, granular glomerular capillary loop deposits are present).

Electron microscopy: Noncontributory (unless associated with membranous nephropathy, in which case, subepithelial glomerular capillary loop deposits are present).

Etiology/Pathogenesis

Sarcoidosis is of unknown etiology and pathogenesis. Abnormal T-cell function, and a possible contribution of infection (eg, atypical mycobacteria), have been postulated. Increased 1,25-dihydroxyvitamin D₃ from macrophages is thought to result in hypercalcemia.

Differential Diagnosis

Noncaseating granulomas are sometimes present with drug-induced hypersensitivity reactions, but are vaguely formed, and usually not numerous and confluent. Granulomas with central necrosis (caseating) are present with infection, such as fungus or tuberculosis, and special stains/culture/polymerase chain reaction amplification for these organisms then typically give positive results. Vague granulomas with fibrinoid necrosis, often involving extensively destroyed glomeruli and/or vascular structures, can be seen in pauci-immune necrotizing crescentic glomerulonephritis, and granulomas can also be encountered in anti-glomerular basement membrane antibody-mediated disease, usually surrounding glomeruli with crescents and necrosis.

Key Diagnostic Features

- Well-formed noncaseating granulomas in the interstitium
- Granulomas are numerous and confluent

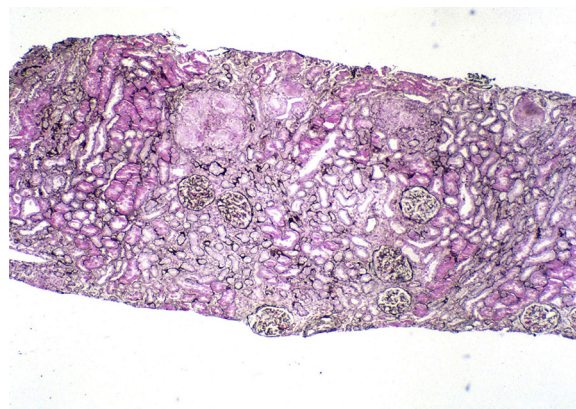


Figure 1. Sarcoidosis in the kidney is characterized by sharply delineated nonnecrotizing granulomas with frequent epithelioid-type multinucleated giant cells with associated interstitial fibrosis and lymphoplasmacytic interstitial infiltrate (Jones stain). Reproduced with permission from AJKD 37(2):e12-e13.

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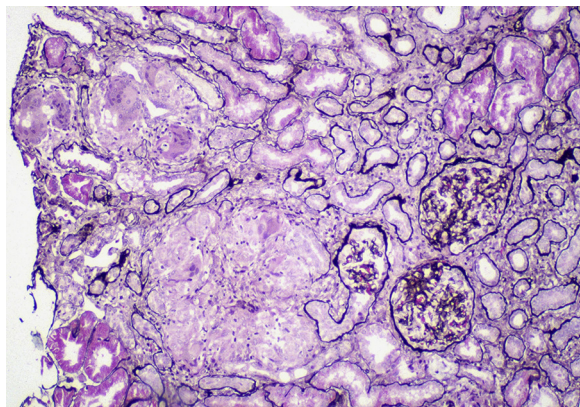


Figure 2. Sarcoidosis in the kidney with well-delineated non-necrotizing granulomas with adjacent tubulointerstitial fibrosis (Jones stain). Reproduced with permission from *AJKD* 37(2):e12-e13.

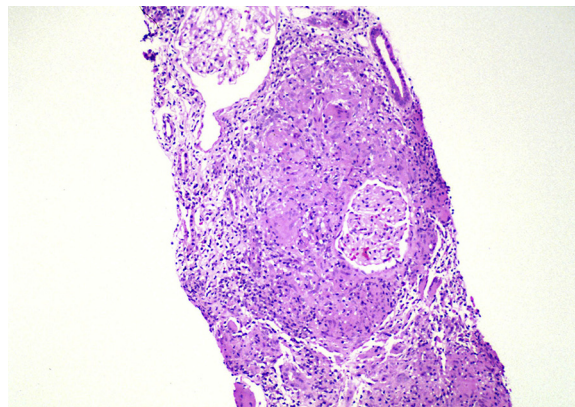


Figure 4. Sarcoidosis in the kidney with well-delineated granuloma with numerous giant cells surrounding an intact glomerulus (hematoxylin and eosin stain). Reproduced with permission from *AJKD* 37(2):e12-e13.

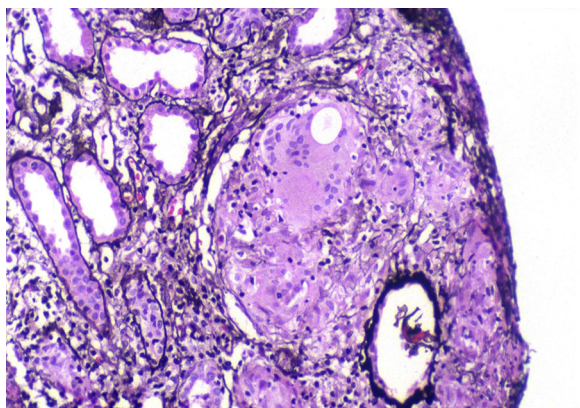


Figure 3. Sarcoidosis in the kidney with nonnecrotizing granulomas comprising large giant cells with surrounding lymphoplasmacytic interstitial infiltrate and early tubulointerstitial fibrosis (Jones stain). Reproduced with permission from *AJKD* 37(2):e12-e13.

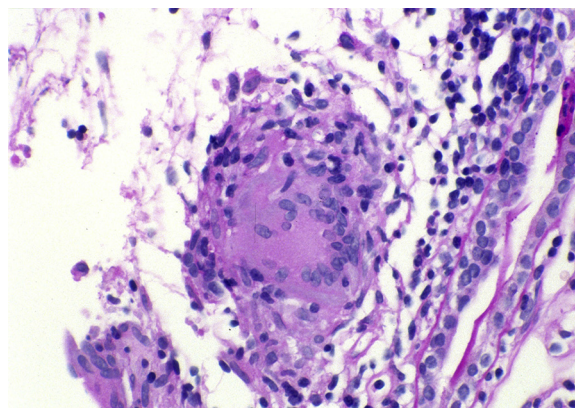


Figure 5. Sarcoidosis in the kidney, with multinucleated giant cells (periodic acid–Schiff stain). Reproduced with permission from *AJKD* 37(2):e12-e13.